

Bundesliga Match Outcome Prediction using CRISP-DM (2019-20 to 2024-25)

References and data provenance

Data source: Football-Data.co.uk, German Bundesliga season CSV files.

Downloads: <https://www.football-data.co.uk/germanym.php>

Column definitions: <https://www.football-data.co.uk/notes.txt>

Advanced model documentation: XGBoost, <https://xgboost.readthedocs.io/en/stable/>

Files used (seasons):

- *bundesliga_19-20.csv*
- *bundesliga_20-21.csv*
- *bundesliga_21-22.csv*
- *bundesliga_22-23.csv*
- *bundesliga_23-24.csv*
- *bundesliga_24-25.csv*

1) Business Understanding

Problem statement

The objective is to predict Bundesliga match outcomes using pre-match information derived from historical team performance. The outcome is modeled as a three-class target:

- HomeWin
- Draw
- AwayWin

The model outputs probabilities for each class, not just a hard prediction.

Sports-related objective and stakeholders

The primary stakeholder is a club analyst who supports the coaching staff and performance departments. A consistent probabilistic baseline can be used for:

- pre-match expectation setting,
- identifying high-risk fixtures (upset potential),
- supporting scenario planning and match preparation with a repeatable quantitative reference.

This is not intended to replace tactical analysis. It is a structured baseline that can be compared to expert judgment.

Hypothesis

Differences in recent form between the home and away teams contain a predictive signal for match outcomes. In particular, rolling points and goal difference trends should correlate with win probability.

Success criteria

A successful first iteration must:

1. Beat naive baselines on a future season hold-out (24-25), demonstrating true out-of-sample value.
2. Use time-aware validation to avoid unrealistic random shuffling.
3. Evaluate probability quality (log loss, Brier score, calibration), because the output is probabilistic and intended for decision support.
4. Provide a clear next iteration plan if performance is limited.

Risks and assumptions

- Football outcomes are noisy and low-scoring, which reduces the maximum achievable predictive performance, especially for draws.
- Team-level match aggregates omit critical context such as injuries, suspensions, lineup selection, fatigue, travel, and tactical matchups.
- Any feature that directly uses post-match statistics would introduce leakage and invalidate conclusions, so strict leakage prevention is required.

2) Data Understanding

Data description

The dataset consists of match-level Bundesliga data across seasons 2019-20 through 2024-25. Core fields include:

- match date and time,
- home team and away team identifiers,
- full-time home goals (FTHG) and away goals (FTAG),
- full-time result indicator (FTR: H, D, A),
- additional match statistics in some seasons.

The target variable is derived from FTR, mapped to HomeWin, Draw, and AwayWin.

Data quality checks and findings

The following checks were performed:

- **Match counts by season:** verified that each season file contributes a consistent number of fixtures (Each season contains 306 fixtures in the dataset).
- **Missingness:** computed missing rate per column.
- **Duplicates:** checked for duplicates using season, date, home team, and away team as key fields.
- **Team naming consistency:** compared unique home vs away team names to ensure symmetry (a mismatch could indicate inconsistent naming or encoding).

Outcomes of checks:

- **Match counts by season:** all seasons contain 306 fixtures (19-20 through 24-25).
- **Missingness** in core fields (season, datetime, HomeTeam, AwayTeam, FTHG, FTAG, FTR): 0%.
- **Duplicate rows** (season, datetime, HomeTeam, AwayTeam key): 0.
- **Team naming consistency:** no mismatches between HomeTeam and AwayTeam lists.

Exploratory data analysis

EDA focused on patterns relevant to football outcomes:

- **Outcome distribution:** HomeWins occur most frequently; Draws are the smallest class; AwayWins are intermediate. This confirms class imbalance.

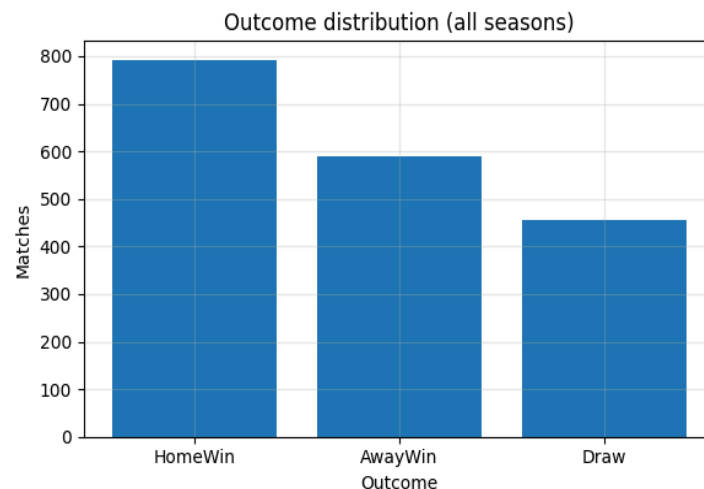


Fig. 1: Outcome distribution (overall bar chart)

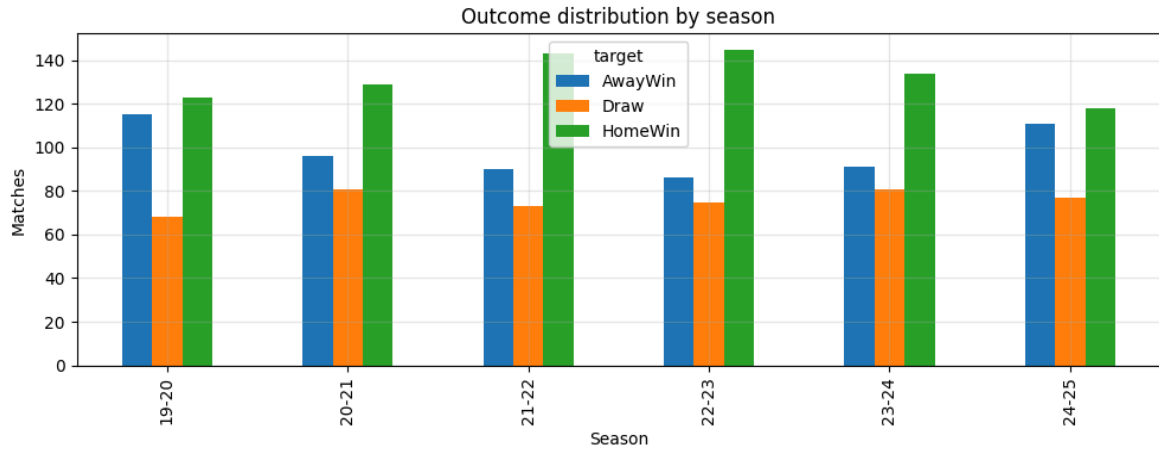


Fig. 2: Outcome distribution by season (grouped bar)

- **Goals distribution:** total goals per match show a skewed distribution typical of football, with most games having between 1 and 4 total goals.

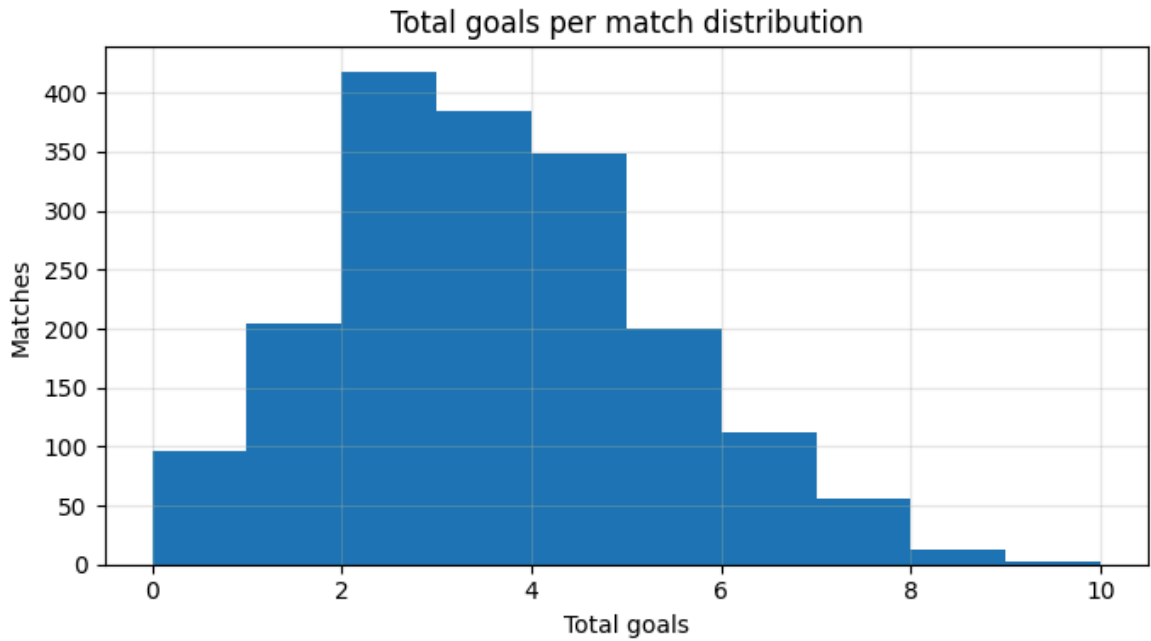


Fig. 3: Total goals histogram

- **Home advantage:** computed average points per match for home vs away teams by season, showing a consistent home advantage pattern, though strength can vary by season.

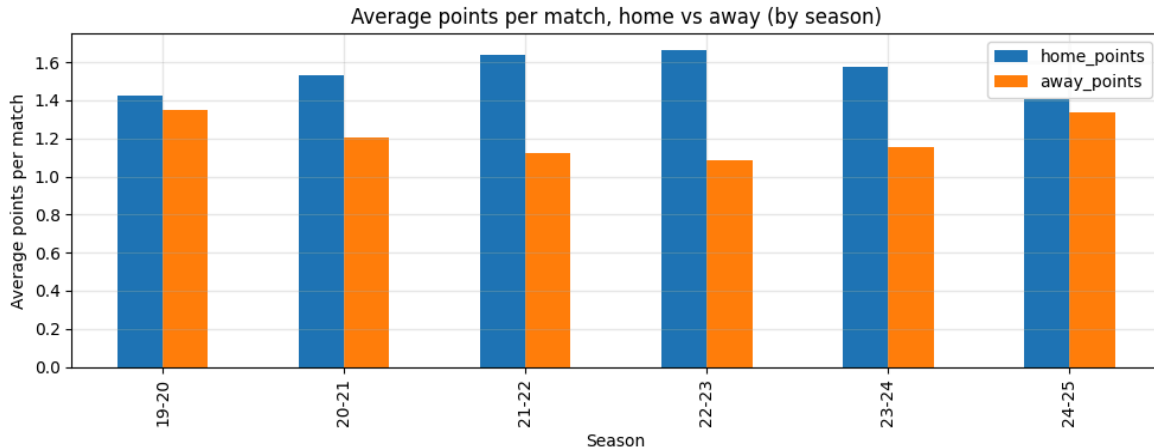


Fig. 4: Home vs away average points per match (by season)

Key observations and implications

- **Class imbalance** suggests accuracy alone is misleading. Macro F1 is needed to evaluate each class fairly.
- **Draws are difficult** by nature; small tactical or random events can shift the match state. A model based only on team-level aggregates will likely misclassify draws.
- **Time structure matters:** football seasons are chronological, and training must respect this to avoid training on “future” information.

Leakage risks identified

Potential leakage could occur if using features available only after the match, such as in-match shots, corners, or cards from the target match. Therefore:

- Only features derived from matches that occurred before the fixture are allowed.
- All feature engineering must be computed sequentially and updated only after each match.

3) Data Preparation

What was done

A leakage-safe feature engineering pipeline was implemented:

1. Sort matches chronologically by datetime.
2. For each match, compute features for home and away teams using only their prior matches in the same season.
3. Update each team's history only after features are created for that match.
4. Reset team histories at season boundaries to avoid ambiguous cross-season carryover.

Feature engineering design

Features were engineered from team histories as:

- **Rolling averages over the last N matches** (N set to 5 in the main model):
 - average points per match,
 - average goals for and goals against,
 - average goal difference.
- **Season-to-date averages:**
 - points per match,
 - goal difference.
- **Home minus away differences:**
 - difference in rolling points, rolling goal difference, and season-to-date equivalents.

These features reflect the “form gap” between teams, aligning naturally with the match-up structure.

Why these choices are reasonable:

- Rolling windows approximate current form, which is actionable pre-match.
- Season-to-date features provide a broader signal of stable team quality.
- Differences (home minus away) directly express the matchup imbalance.

Data splitting strategy

To simulate real deployment:

- Training: seasons 2019-20 through 2023-24
- Testing: season 2024-25 (future hold-out)

This design ensures that the evaluation reflects performance on unseen future data.

Limitations of the preparation step

- Resetting history each season may remove real carryover effects, but it avoids ambiguity and is a conservative leakage-safe choice.
- Team-level aggregates cannot capture player-specific shocks (injuries, transfers, coaching changes), reducing predictive ceiling.
- Feature engineering is limited to what can be derived from match results and therefore, cannot fully explain tactical variance.

4) Modeling

Modeling approach

We compared multiple methods to ensure the modeling phase is rigorous:

Baseline 1: Naive strategies

- Always predict HomeWin.
- Always predict the majority class.

These are important for establishing a minimum performance threshold.

Baseline 2: Multinomial Logistic Regression

A strong and interpretable baseline that:

- handles multi-class outcomes directly,
- provides probability outputs,
- works well when the relationship between features and outcome is approximately linear.

index	model	accuracy	macro_f1	log_loss	brier
0	always_home	0.385621	0.185535	22.968995	1.228758
1	majority	0.385621	0.185535	22.968995	1.228758
2	logreg	0.395425	0.376237	1.251210	0.641540
3	xgboost	0.450980	0.380286	1.372663	0.672071

Table 1: Model comparison on hold-out (Accuracy, Macro F1, Log loss, Brier)

Advanced model: XGBoost

XGBoost is a gradient boosted decision tree model designed for tabular prediction tasks. It is often used in applied ML and competitions because it can model non-linear feature interactions and typically performs strongly on structured data.

Documentation: <https://xgboost.readthedocs.io/en/stable/>

Hyperparameter tuning and experimentation

Hyperparameter tuning is essential for a fair evaluation of advanced models. We used:

- **GridSearchCV** for systematic exploration of hyperparameters,
- **TimeSeriesSplit** to preserve chronological order during cross-validation.

Parameters tuned included:

- max_depth
- learning_rate
- n_estimators
- subsample
- colsample_bytree

This satisfies the requirement to experiment with methods in the same model family and to tune using GridSearch.

Why time-aware validation is critical

Random k-fold splits would mix future matches into training folds, inflating performance. TimeSeriesSplit is a conservative and realistic approach that mimics deployment, where future games are unknown.

Model selection criterion

Macro F1 was used as the primary selection metric because it treats all outcome classes equally under imbalance. Log loss and Brier score were retained as probability-quality indicators.

5) Evaluation

Evaluation setup

Final evaluation was performed on the held-out 2024-25 season. This is the most important part of the methodology because it determines whether the model answers the initial question and whether it generalizes.

Metrics used and justification

- **Accuracy:** easy to interpret, but can be misleading with imbalance.
- **Macro F1:** assigns equal weight to each class; important when draws are rare.
- **Log loss:** evaluates probability quality and penalizes overconfident, incorrect predictions.
- **Multi-class Brier score:** measures probability calibration quality in an intuitive squared-error sense.
- **Confusion matrix:** shows which outcomes are misclassified, crucial for diagnosing draw performance.
- **Calibration curves:** assess whether predicted probabilities are reliable.

Results summary (hold-out season 24-25)

General pattern observed in this project setup:

- Naive baselines perform poorly, especially on Macro F1 and probability metrics, confirming that a simple heuristic is insufficient.
- XGBoost slightly improves accuracy and Macro F1 in this run, but it does not improve probability metrics (log loss, Brier) compared to Logistic Regression.
- Logistic Regression provides the best probability quality and a competitive Macro F1, indicating that with form-gap features, the relationship may be close to linear.

We select Logistic Regression as the final model for Figures 5 and 6 because it yields better probability quality (lower log loss and Brier) and is a strong, interpretable baseline, whereas XGBoost is reported as the slightly higher Macro F1 alternative in this run.

Critical analysis

Hypothesis: Form differences contain a predictive signal.

Answer: Yes, partially.

- Models beat naive baselines, confirming that historical form features do contain predictive signal.
- However, overall performance remains moderate, and draw predictions remain challenging, indicating that form aggregates alone are insufficient to robustly separate all three classes.

This is an important scientific conclusion: the limiting factor appears to be missing context features, not necessarily the model's complexity.

Main failure mode: draws

Draws are typically the hardest class because:

- Football is low-scoring; one event can flip the result class.
- Form aggregates do not capture tactical matchups and lineup-specific constraints.
- Draws may represent an “in-between” state, requiring richer features (e.g., expected goals, match tempo, shot-quality proxies) that are not available pre-match in this dataset.

The confusion matrix typically shows many draws being predicted as narrow home wins or away wins.

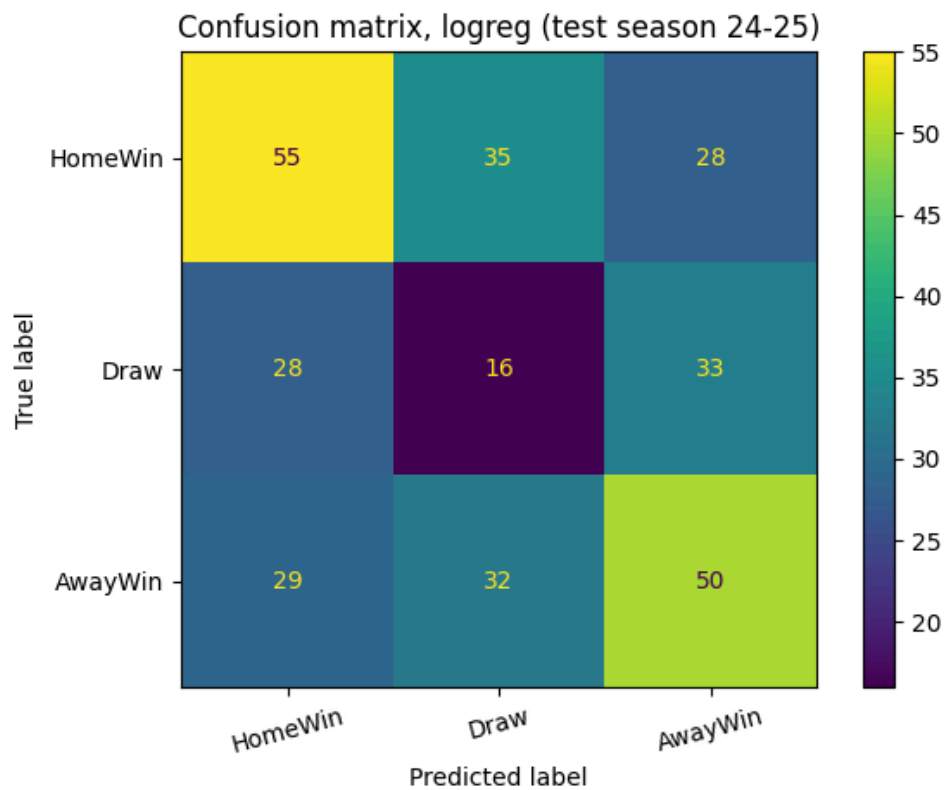


Fig. 5: Confusion matrix of best model on 2024-25 (absolute values)

Class-wise metrics confirm that draw recall is the lowest (Draw recall = 0.208, Draw F1 = 0.200), consistent with the confusion matrix. This indicates that form-gap features capture win tendencies better than the “in-between” draw state, which likely requires richer context features.

Per-team error analysis

A per-team analysis was conducted using negative log likelihood assigned to both teams per match. This helps identify:

- teams for which model probabilities are consistently poor,
- potential structural reasons (high variance teams, style of play, mid-season changes).

This is valuable because a model can have average performance but systematically fail on specific teams, which matters to clubs.

Reliability of probabilities

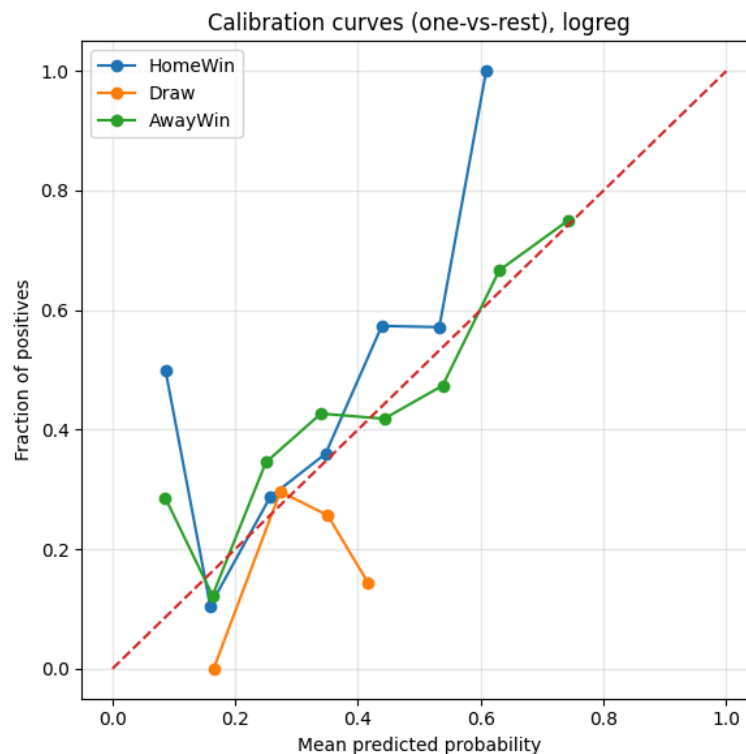


Fig. 6: Calibration curves (one-vs-rest) of best model

Calibration curves provide an additional check:

- if the model is overconfident, it can mislead decision-making even when accuracy is acceptable,
- probability calibration matters when the output is used for scenario planning.

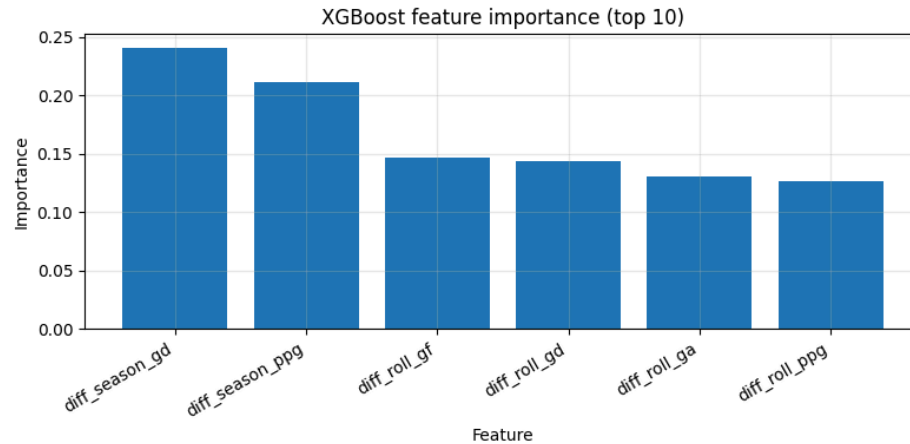


Fig. 7: Feature importance (XGBoost) or coefficients (LogReg)

XGBoost is included to test whether non-linear interactions among form-gap features improve generalization beyond a linear baseline.

What should change in the next CRISP-DM iteration

Because the hypothesis is only partially supported and draw prediction remains weak, the next iteration should focus on:

1. **Add schedule and fatigue features:** rest days, travel, midweek match flags, congestion indicators.
2. **Add player availability:** injuries, suspensions, lineup stability and minutes load.
3. **Add team strength priors:** Elo ratings or matchday ranking features to represent baseline quality.
4. **Consider alternative targets aligned to decisions:** for example, avoid defeat vs lose, which is often more actionable for coaching decisions and easier to predict reliably.
5. **Evaluate across multiple hold-out seasons:** if available, to ensure conclusions are not specific to one season.

Final conclusion

This CRISP-DM project demonstrates a rigorous and replicable ML workflow in a sports context. The key contribution is not only model training but an honest evaluation on a future season hold-out with probability-focused metrics and critical analysis.

The results show that:

- The team form contains a real but limited predictive signal,
- Naive strategies are clearly inferior,
- Advanced models like XGBoost are valuable to test, but without richer features, they may not outperform simpler baselines on class-based metrics,
- probability evaluation and calibration provide deeper insight into model reliability,
- The next iteration should prioritize adding pre-match context and decision-aligned targets to increase practical utility.